<u>CCNA</u>

<u>VLAN and INTER-VLAN Routing TASK – 2</u>

1. **In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.**

   Ans :

Branch Office A	Branch Office B
PC-A1	PC-B1
192.168.1.10	192.168.2.10
|	|
|	|
[Switch A]	[Switch B]
|	|
|	|
[Router R1]	[Router R2]
192.168.1.1	192.168.2.1

2. **Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.**

Ans :

```
Branch A                          Branch B
PC-A1                            PC-B1
192.168.1.10                        192.168.2.10
   |                             |
 Switch A                         Switch B
   |                             |
 R1                             R2
G0/0: 192.168.1.1                  G0/0: 192.168.2.1
G0/1: 10.0.0.1                   G0/1: 10.0.0.2
   \                             /
    \                           /
     ======= GRE Tunnel 0 ====================
      172.16.0.1       172.16.0.2
```

3. **Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.

Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.**

Ans :

```
Remote User
   PC
192.168.10.10
    |
```

```
     |  VPN / Internet
     |
     v
   R1 VPN Server
  10.0.0.1
     |
     | LAN
     |
  Switch
     |
 Branch PC
192.168.1.10
```

4. **List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.**

   Ans :

Feature	**Site-to-Site VPN**	**Remote Access VPN**
Purpose	Connects two entire branch networks.	Connects an individual user/device to a branch network.
Configuration	Configured mainly on **both routers** using a GRE tunnel.	Configured on the **VPN server/router**

Feature	Site-to-Site VPN	Remote Access VPN
		and client PC using username/password authentication.
Users	Many users at both branches can communicate through the VPN.	Usually one remote user/device connects at a time.
Example from Packet Tracer	R1 ↔ R2 using Tunnel 0 with 172.16.0.1 and 172.16.0.2.	Remote PC → R1 using the PC's VPN tab.
Best suited for	Permanent branch-to-branch connectivity.	Employees working from home or traveling.

Site-to-Site VPN

Advantages:

- **Connects entire networks:** All authorized devices in both branches can communicate without configuring a VPN separately on every PC.
- **Good for permanent connections:** Once configured, the tunnel can provide continuous connectivity between branch offices.

Disadvantages:

- **More complex setup:** Both routers need appropriate tunnel, routing, and VPN configuration.
- **Less flexible for individual users:** It is not ideal when employees need to connect from changing locations.

Remote Access VPN

Advantages:

- **Supports remote workers:** An employee can connect to the office network from home or another location.
- **Individual authentication:** Users can be given separate usernames and passwords, making access easier to control.

Disadvantages:

- **Client configuration is required:** Each user's PC needs to be configured with the VPN server address and login credentials.
- **Dependent on the user's Internet connection:** A poor or unstable connection can affect access to company resources.

5. **Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.**

Ans :

Device/Interface	IP Address	Purpose
R1 G0/0	192.168.1.1/24	Branch A LAN
R1 G0/1	10.0.0.1/30	WAN connection
R1 Tunnel 0	172.16.0.1/30	GRE tunnel
R2 G0/0	192.168.2.1/24	Branch B LAN
R2 G0/1	10.0.0.2/30	WAN connection

Device/Interface	IP Address	Purpose
R2 Tunnel 0	172.16.0.2/30	GRE tunnel